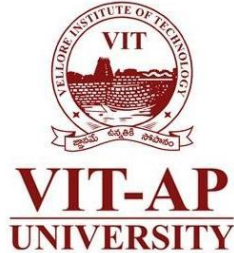


MAT1001 Calculus for Engineers

Surface Integral, Divergence Theorem and Stokes' Theorem



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Formulas for a Surface Integral

1. For a smooth surface S defined **parametrically** as $\mathbf{r}(u, v) = f(u, v)\mathbf{i} + g(u, v)\mathbf{j} + h(u, v)\mathbf{k}$, $(u, v) \in R$, and a continuous function $G(x, y, z)$ defined on S , the surface integral of G over S is given by the double integral over R ,

$$\iint_S G(x, y, z) d\sigma = \iint_R G(f(u, v), g(u, v), h(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| du dv. \quad (2)$$

2. For a surface S given **implicitly** by $F(x, y, z) = c$, where F is a continuously differentiable function, with S lying above its closed and bounded shadow region R in the coordinate plane beneath it, the surface integral of the continuous function G over S is given by the double integral over R ,

$$\iint_S G(x, y, z) d\sigma = \iint_R G(x, y, z) \frac{|\nabla F|}{|\nabla F \cdot \mathbf{p}|} dA, \quad (3)$$

where $\mathbf{p}$ is a unit vector normal to R and $\nabla F \cdot \mathbf{p} \neq 0$.

3. For a surface S given **explicitly** as the graph of $z = f(x, y)$, where f is a continuously differentiable function over a region R in the xy -plane, the surface integral of the continuous function G over S is given by the double integral over R ,

$$\iint_S G(x, y, z) d\sigma = \iint_R G(x, y, f(x, y)) \sqrt{f_x^2 + f_y^2 + 1} dx dy. \quad (4)$$

Example

• Integrate $G(x, y, z) = x^2$ over the cone $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq 1$.

→

$$\mathbf{r}(r, \theta) = (r \cos \theta)\mathbf{i} + (r \sin \theta)\mathbf{j} + r\mathbf{k}, \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

$$|\mathbf{r}_r \times \mathbf{r}_\theta| = \sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta + r^2} = \sqrt{2r^2} = \sqrt{2}r.$$

$$\iint_S x^2 d\sigma = \int_0^{2\pi} \int_0^1 (r^2 \cos^2 \theta)(\sqrt{2}r) dr d\theta \quad x = r \cos \theta$$

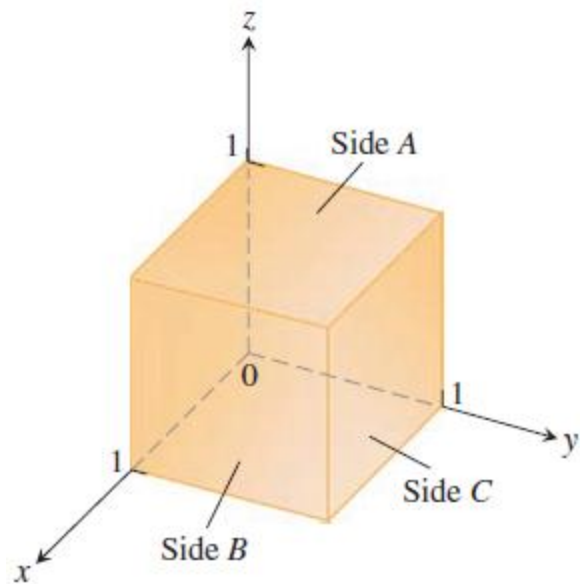
$$= \sqrt{2} \int_0^{2\pi} \int_0^1 r^3 \cos^2 \theta dr d\theta$$

$$= \frac{\sqrt{2}}{4} \int_0^{2\pi} \cos^2 \theta d\theta = \frac{\sqrt{2}}{4} \left[\frac{\theta}{2} + \frac{1}{4} \sin 2\theta \right]_0^{2\pi} = \frac{\pi\sqrt{2}}{4}.$$

Example

- Integrate $G(x, y, z) = xyz$ over the surface of the cube cut from the first octant by the planes

$$x = 1, y = 1, \text{ and } z = 1$$



We integrate xyz over each of the six sides and add the results.

Since $xyz = 0$ on the sides that lie in the coordinate planes,

$$\iint_{\text{Cube surface}} xyz \, d\sigma = \iint_{\text{Side A}} xyz \, d\sigma + \iint_{\text{Side B}} xyz \, d\sigma + \iint_{\text{Side C}} xyz \, d\sigma.$$

Example

● Integrate $G(x, y, z) = xyz$ over the surface of the cube cut from the first octant by the planes

$$x = 1, y = 1, \text{ and } z = 1$$



Side A is the surface $f(x, y, z) = z = 1$ over the square region

$$R_{xy}: 0 \leq x \leq 1, 0 \leq y \leq 1, \text{ in the } xy\text{-plane.}$$

$$\mathbf{p} = \mathbf{k}, \quad \nabla f = \mathbf{k}, \quad |\nabla f| = 1, \quad |\nabla f \cdot \mathbf{p}| = |\mathbf{k} \cdot \mathbf{k}| = 1$$

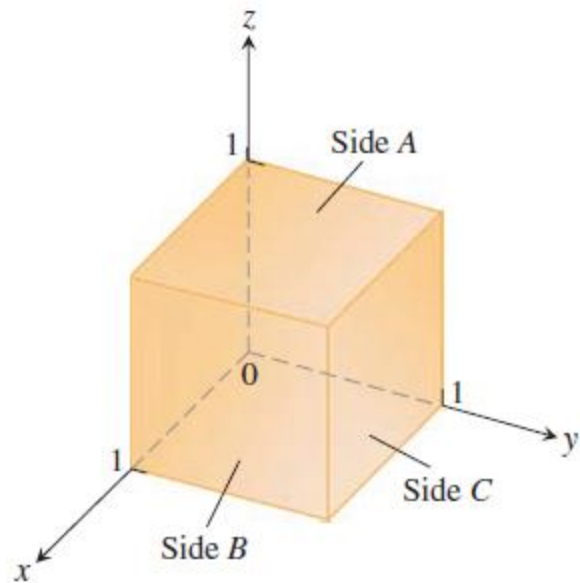
$$d\sigma = \frac{|\nabla f|}{|\nabla f \cdot \mathbf{p}|} dA = \frac{1}{1} dx dy = dx dy$$

$$xyz = xy(1) = xy$$

$$\iint_{\text{Side A}} xyz d\sigma = \iint_{R_{xy}} xy dx dy = \int_0^1 \int_0^1 xy dx dy = \int_0^1 \frac{y}{2} dy = \frac{1}{4}.$$

integrals of xyz over sides B and C are also $1/4$. Hence,

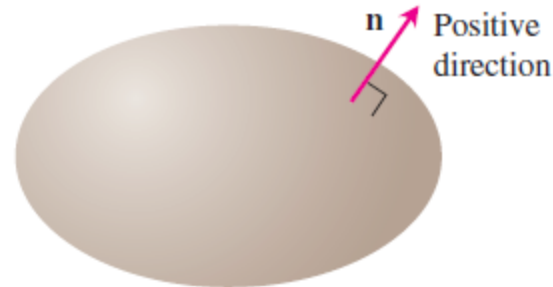
$$\iint_{\text{Cube surface}} xyz d\sigma = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}.$$



Orientation of Surfaces

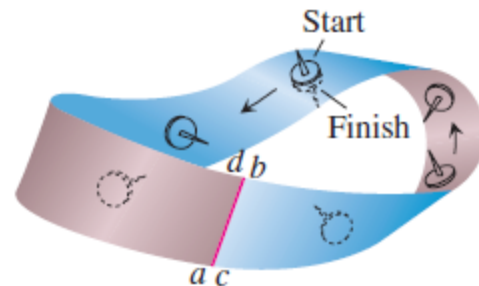
We call a smooth surface S **orientable** or **two-sided** if it is possible to define a field $\mathbf{n}$ of unit normal vectors on S that varies continuously with position. Any patch or subportion of an orientable surface is orientable.

By convention, we choose $\mathbf{n}$ on a closed surface to point outward.



Once $\mathbf{n}$ has been chosen, we say that we have **oriented** the surface, and we call the surface together with its normal field an **oriented surface**. The vector $\mathbf{n}$ at any point is called the **positive direction** at that point (Figure 16.49).

The Möbius band is not orientable.



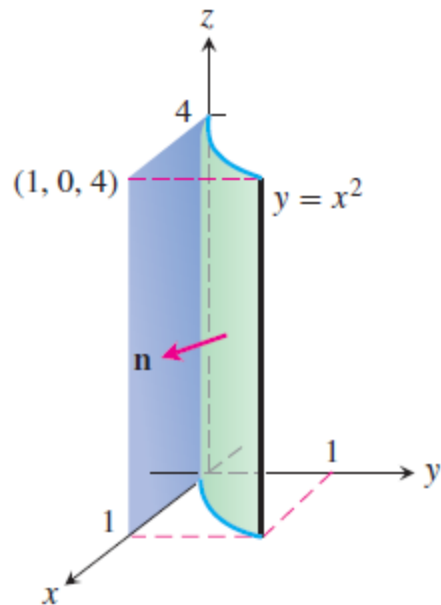
DEFINITION The **flux** of a three-dimensional vector field **F** across an oriented surface S in the direction of **n** is

$$\text{Flux} = \iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma. \quad (5)$$

Example

● Find the flux of $\mathbf{F} = yz\mathbf{i} + x\mathbf{j} - z^2\mathbf{k}$ through the parabolic cylinder $y = x^2$, $0 \leq x \leq 1$, $0 \leq z \leq 4$.

→



On the surface we have $x = x$, $y = x^2$, and $z = z$, so we have the parametrization

$$\mathbf{r}(x, z) = x\mathbf{i} + x^2\mathbf{j} + z\mathbf{k}, \quad 0 \leq x \leq 1, \quad 0 \leq z \leq 4.$$

$$\mathbf{r}_x \times \mathbf{r}_z = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2x & 0 \\ 0 & 0 & 1 \end{vmatrix} = 2x\mathbf{i} - \mathbf{j}.$$

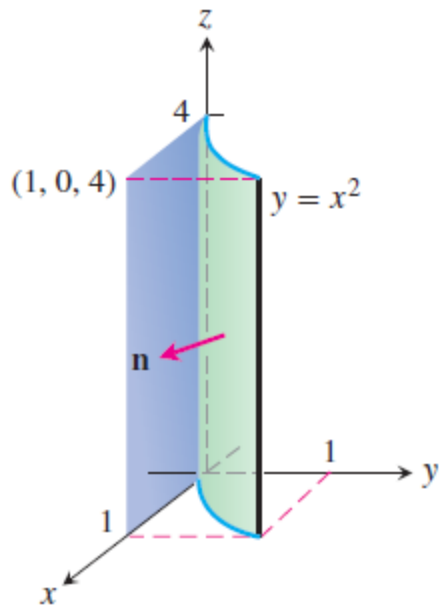
$$\mathbf{n} = \frac{\mathbf{r}_x \times \mathbf{r}_z}{|\mathbf{r}_x \times \mathbf{r}_z|} = \frac{2x\mathbf{i} - \mathbf{j}}{\sqrt{4x^2 + 1}}.$$

On the surface, $y = x^2$,

$$\mathbf{F} = yz\mathbf{i} + x\mathbf{j} - z^2\mathbf{k} = x^2z\mathbf{i} + x\mathbf{j} - z^2\mathbf{k}.$$

Example

Find the flux of $\mathbf{F} = yz\mathbf{i} + x\mathbf{j} - z^2\mathbf{k}$ through the parabolic cylinder $y = x^2$, $0 \leq x \leq 1$, $0 \leq z \leq 4$.



$$\mathbf{n} = \frac{\mathbf{r}_x \times \mathbf{r}_z}{|\mathbf{r}_x \times \mathbf{r}_z|} = \frac{2x\mathbf{i} - \mathbf{j}}{\sqrt{4x^2 + 1}}.$$

$$\mathbf{F} = yz\mathbf{i} + x\mathbf{j} - z^2\mathbf{k} = x^2z\mathbf{i} + x\mathbf{j} - z^2\mathbf{k}.$$

$$\begin{aligned} \mathbf{F} \cdot \mathbf{n} &= \frac{1}{\sqrt{4x^2 + 1}} ((x^2z)(2x) + (x)(-1) + (-z^2)(0)) \\ &= \frac{2x^3z - x}{\sqrt{4x^2 + 1}}. \end{aligned}$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma &= \int_0^4 \int_0^1 \frac{2x^3z - x}{\sqrt{4x^2 + 1}} |\mathbf{r}_x \times \mathbf{r}_z| \, dx \, dz \\ &= \int_0^4 \int_0^1 \frac{2x^3z - x}{\sqrt{4x^2 + 1}} \sqrt{4x^2 + 1} \, dx \, dz = 2. \end{aligned}$$

Implicit Surfaces and Flux

If S is part of a level surface $g(x, y, z) = c$, then $\mathbf{n}$ may be taken to be one of the two fields

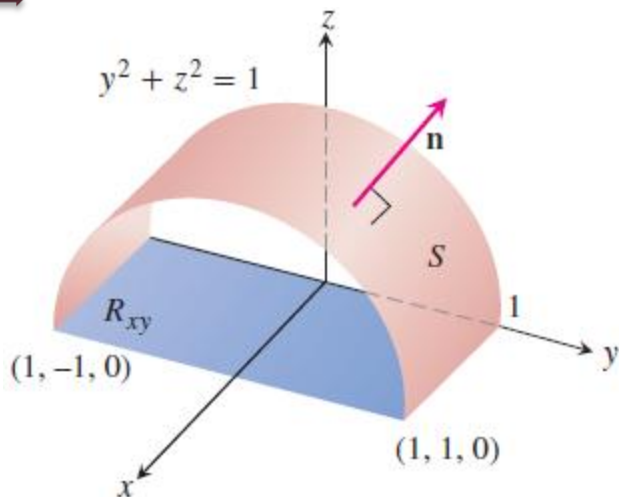
$$\mathbf{n} = \pm \frac{\nabla g}{|\nabla g|},$$

depending on which one gives the preferred direction. The corresponding flux is

$$\begin{aligned}\text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma \\ &= \iint_R \left(\mathbf{F} \cdot \frac{\pm \nabla g}{|\nabla g|} \right) \frac{|\nabla g|}{|\nabla g \cdot \mathbf{p}|} \, dA \\ &= \iint_R \mathbf{F} \cdot \frac{\pm \nabla g}{|\nabla g \cdot \mathbf{p}|} \, dA.\end{aligned}$$

Example

- Find the flux of $\mathbf{F} = yz\mathbf{j} + z^2\mathbf{k}$ outward through the surface S cut from the cylinder $y^2 + z^2 = 1$, $z \geq 0$, by the planes $x = 0$ and $x = 1$.



$$g(x, y, z) = y^2 + z^2$$

$$\mathbf{n} = +\frac{\nabla g}{|\nabla g|} = \frac{2y\mathbf{j} + 2z\mathbf{k}}{\sqrt{4y^2 + 4z^2}} = \frac{2y\mathbf{j} + 2z\mathbf{k}}{2\sqrt{1}} = y\mathbf{j} + z\mathbf{k}.$$

$$\mathbf{p} = \mathbf{k}, \quad d\sigma = \frac{|\nabla g|}{|\nabla g \cdot \mathbf{k}|} dA = \frac{2}{|2z|} dA = \frac{1}{z} dA.$$

$$\begin{aligned} \mathbf{F} \cdot \mathbf{n} &= (yz\mathbf{j} + z^2\mathbf{k}) \cdot (y\mathbf{j} + z\mathbf{k}) \\ &= y^2z + z^3 = z(y^2 + z^2) \\ &= z. \end{aligned}$$

$$\iint_S \mathbf{F} \cdot \mathbf{n} d\sigma = \iint_S (z) \left(\frac{1}{z} dA \right) = \iint_{R_{xy}} dA = \text{area}(R_{xy}) = 2.$$

Divergence Theorem

THEOREM (*The Divergence Theorem*) *Let G be a solid whose surface σ is oriented outward. If*

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

where f , g , and h have continuous first partial derivatives on some open set containing G , and if $\mathbf{n}$ is the outward unit normal on σ , then

$$\iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G \operatorname{div} \mathbf{F} \, dV \quad (1)$$

Divergence Theorem or Gauss's Theorem

The outward flux of a vector field across a closed surface is equal to the triple integral of the divergence over the region enclosed by the surface.

Example

- Use the Divergence Theorem to find the outward flux of the vector field $\mathbf{F}(x, y, z) = z\mathbf{k}$ across the sphere

$$x^2 + y^2 + z^2 = a^2.$$



$$\mathbf{F}(x, y, z) = z\mathbf{k}$$

$$\text{div } \mathbf{F} = \frac{\partial z}{\partial z} = 1$$

$$\Phi = \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G dV = \text{volume of } G = \frac{4\pi a^3}{3}$$

Example

- Use the Divergence Theorem to find the outward flux of the vector field

$$\mathbf{F}(x, y, z) = 2x\mathbf{i} + 3y\mathbf{j} + z^2\mathbf{k}$$

across the unit cube in Figure 15.7.3.

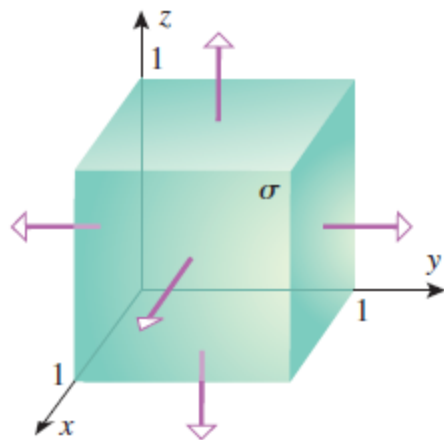


Figure 15.7.3

$$\mathbf{F}(x, y, z) = 2x\mathbf{i} + 3y\mathbf{j} + z^2\mathbf{k}$$

$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(3y) + \frac{\partial}{\partial z}(z^2) = 5 + 2z$$

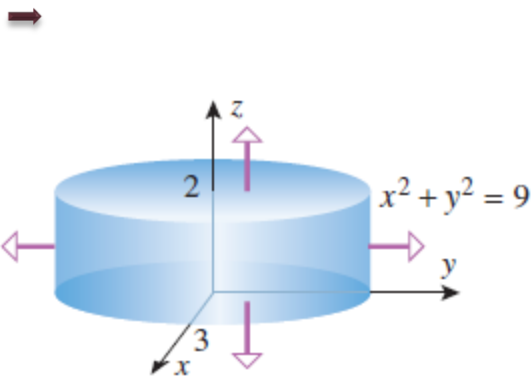
$$\begin{aligned} \Phi &= \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_G (5 + 2z) \, dV = \int_0^1 \int_0^1 \int_0^1 (5 + 2z) \, dz \, dy \, dx \\ &= \int_0^1 \int_0^1 [5z + z^2]_{z=0}^1 \, dy \, dx = \int_0^1 \int_0^1 6 \, dy \, dx = 6 \end{aligned}$$

Example

- Use the Divergence Theorem to find the outward flux of the vector field

$$\mathbf{F}(x, y, z) = x^3\mathbf{i} + y^3\mathbf{j} + z^2\mathbf{k}$$

across the surface of the region enclosed by the circular cylinder $x^2 + y^2 = 9$ and the planes $z = 0$ and $z = 2$



$$\mathbf{F}(x, y, z) = x^3\mathbf{i} + y^3\mathbf{j} + z^2\mathbf{k}$$

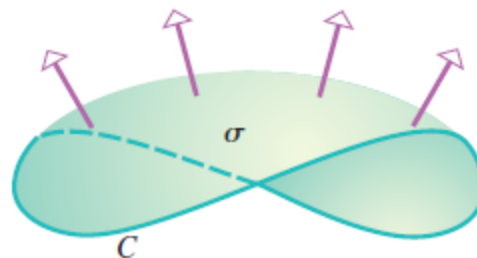
$$\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^2) = 3x^2 + 3y^2 + 2z$$

$$\Phi = \iiint_{\sigma} \mathbf{F} \cdot \mathbf{n} dS = \iiint_G (3x^2 + 3y^2 + 2z) dV$$

$$= \int_0^{2\pi} \int_0^3 \int_0^2 (3r^2 + 2z)r dz dr d\theta$$

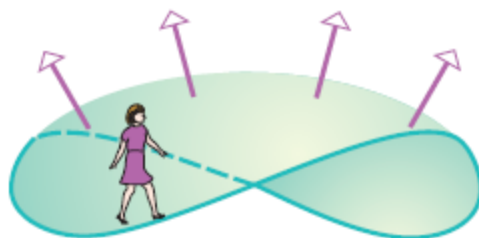
Using cylindrical coordinates

$$= \int_0^{2\pi} \frac{279}{2} d\theta = 279\pi$$

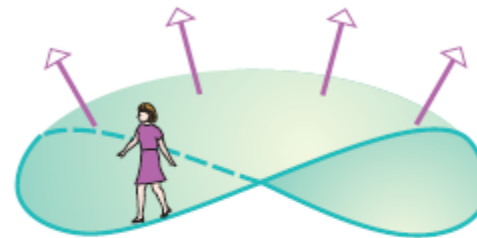


The oriented surface σ is bounded by the simple closed curve C .

If σ is an oriented surface bounded by a simple closed parametric curve C , then there are two possible relationships between the orientations of σ and C , which can be described as follows.



The positive direction of C
relative to the orientation of σ .



The negative direction of C
relative to the orientation of σ .

Imagine a person walking along the curve C with his or her head in the direction of the orientation of σ .

The person is said to be walking in the *positive direction* of C relative to the orientation of σ if the surface is on the person's left, and the person is said to be walking in the *negative direction* of C relative to the orientation of σ if the surface is on the person's right

THEOREM (Stokes' Theorem) *Let σ be a piecewise smooth oriented surface that is bounded by a simple, closed, piecewise smooth curve C with positive orientation. If the components of the vector field*

$$\mathbf{F}(x, y, z) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$$

are continuous and have continuous first partial derivatives on some open set containing σ , and if $\mathbf{T}$ is the unit tangent vector to C , then

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS \quad (2)$$

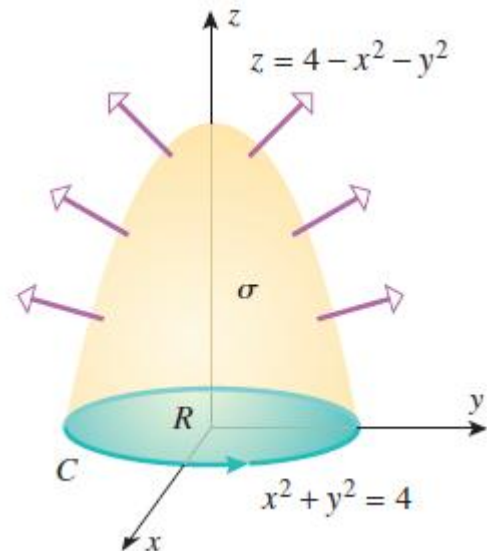
$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix}$$

The work performed by a force field on a particle that traverses a simple, closed, piecewise smooth curve C in the positive direction can be obtained by integrating the normal component of the curl over an oriented surface σ bounded by C .

Example

- Verify Stokes' Theorem for the vector field $\mathbf{F}(x, y, z) = 2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}$, taking σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation, and C to be the positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy -plane



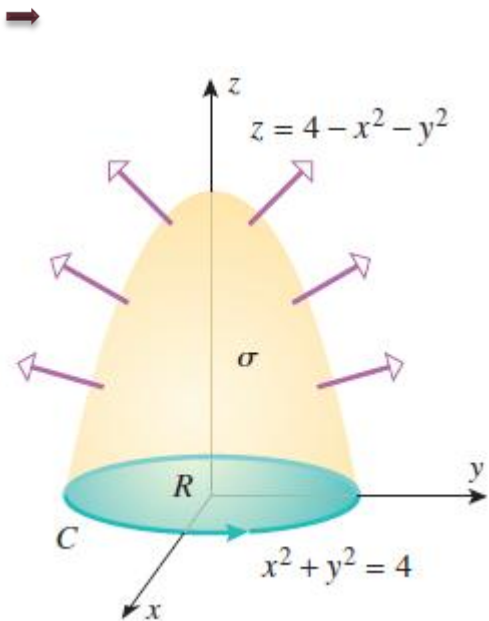
Since σ is oriented up, the positive orientation of C is counterclockwise looking down the positive z -axis. Thus, C can be represented parametrically by

$$x = 2 \cos t, \quad y = 2 \sin t, \quad z = 0 \quad (0 \leq t \leq 2\pi)$$

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C 2z dx + 3x dy + 5y dz \\ &= \int_0^{2\pi} [0 + (6 \cos t)(2 \cos t) + 0] dt \\ &= \int_0^{2\pi} 12 \cos^2 t dt = 12 \left[\frac{1}{2}t + \frac{1}{4} \sin 2t \right]_0^{2\pi} = 12\pi \end{aligned}$$

Example

- Verify Stokes' Theorem for the vector field $\mathbf{F}(x, y, z) = 2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}$, taking σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation, and C to be the positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy -plane



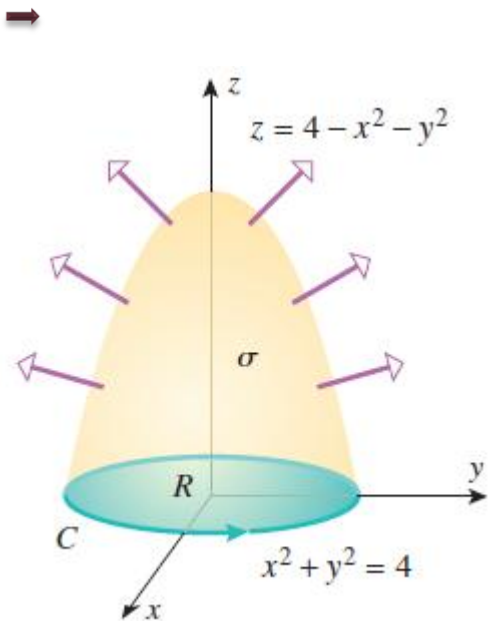
$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z & 3x & 5y \end{vmatrix} = 5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$$

$$z = g(x, y) = 4 - x^2 - y^2$$

$$\begin{aligned} \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \iint_R (\text{curl } \mathbf{F}) \cdot \left(-\frac{\partial z}{\partial x} \mathbf{i} - \frac{\partial z}{\partial y} \mathbf{j} + \mathbf{k} \right) dA \\ &= \iint_R (5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \cdot (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}) dA \end{aligned}$$

Example

- Verify Stokes' Theorem for the vector field $\mathbf{F}(x, y, z) = 2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}$, taking σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation, and C to be the positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy -plane



$$\begin{aligned}
 \iint_{\sigma} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \iint_R (\text{curl } \mathbf{F}) \cdot \left(-\frac{\partial z}{\partial x} \mathbf{i} - \frac{\partial z}{\partial y} \mathbf{j} + \mathbf{k} \right) dA \\
 &= \iint_R (5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) \cdot (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}) dA \\
 &= \iint_R (10x + 4y + 3) dA \\
 &= \int_0^{2\pi} \int_0^2 (10r \cos \theta + 4r \sin \theta + 3)r \, dr \, d\theta = 12\pi
 \end{aligned}$$

Thank You...